

Midterm Exam 1 · Fall 2021

MATH 411 · October 1, 2021

Complete worked solutions

Retypeset past exam for practice. The original was closed book and worth 25 points.

1. Let $a, b, c, d \in \mathbb{R}$ satisfy $ad - bc = 1$, and set

$$F(z) = \frac{az + b}{cz + d}, \quad \mathbb{H} = \{z \in \mathbb{C} : \operatorname{Im} z > 0\}.$$

- (a) **(2 points)** Show that F maps $\mathbb{H}$ into $\mathbb{H}$.

Solution. The denominator cannot vanish on $\mathbb{H}$: if $c \neq 0$ its zero is the real number $-d/c$, while if $c = 0$ then $d \neq 0$. For $z = x + iy$,

$$\operatorname{Im} F(z) = \frac{(ad - bc)y}{|cz + d|^2} = \frac{y}{|cz + d|^2} > 0.$$

- (b) **(2 points)** Is $F : \mathbb{H} \rightarrow \mathbb{H}$ holomorphic?

Solution. Yes. It is a quotient of holomorphic functions, and its denominator has no zero in $\mathbb{H}$.

- (c) **(5 points)** A line or circle can be given by an equation

$$Ax + By + C(x^2 + y^2) = D,$$

where $A, B, C, D \in \mathbb{R}$ and $z = x + iy$. For the choice $(a, b, c, d) = (0, -1, 1, 0)$, show that $F(z) = -1/z$ sends the portion of such a line or circle in $\mathbb{H}$ to the portion of a line or circle in $\mathbb{H}$.

Solution. Write $w = u + iv = -1/z$. As $z = -1/w$ and $w \in \mathbb{H}$ by part (a),

$$x = -\frac{u}{u^2 + v^2}, \quad y = \frac{v}{u^2 + v^2}, \quad x^2 + y^2 = \frac{1}{u^2 + v^2}.$$

Substitution and multiplication by $u^2 + v^2$ give

$$-Au + Bv - D(u^2 + v^2) = -C.$$

This is again an equation of the stated form: a line when $D = 0$, or a circle when $D \neq 0$ (provided the locus is nonempty). Since F is its own inverse, this equation describes precisely the image in $\mathbb{H}$.

2. Consider the real function

$$f(x) = \begin{cases} e^{-1/x^2}, & x \neq 0, \\ 0, & x = 0. \end{cases}$$

- (a) **(3 points)** Show that f is infinitely differentiable at 0 and compute every $f^{(k)}(0)$. You may use that a rational function of x multiplied by e^{-1/x^2} tends to 0 as $x \rightarrow 0$.

Solution. For $x \neq 0$ put $t = 1/x$. Induction gives $f^{(k)}(x) = P_k(t)e^{-t^2}$, with $P_0 = 1$ and

$$P_{k+1}(t) = -t^2 P'_k(t) + 2t^3 P_k(t).$$

Each P_k is a polynomial. Suppose $f^{(k)}(0) = 0$. Then

$$f^{(k+1)}(0) = \lim_{h \rightarrow 0} \frac{f^{(k)}(h) - f^{(k)}(0)}{h} = \lim_{h \rightarrow 0} \frac{P_k(1/h)e^{-1/h^2}}{h} = 0.$$

Starting with $f(0) = 0$ proves that all derivatives exist and $f^{(k)}(0) = 0$ for every $k \geq 0$.

- (b) **(3 points)** Prove or disprove: an analytic function on a complex neighborhood of 0 agrees with f on the real axis near 0.

Solution. No such analytic function exists. Its complex derivatives at 0 would equal the corresponding derivatives of its restriction to the real axis, hence would all be zero by part (a). Its Taylor series would therefore make it identically zero near 0, but $f(x) > 0$ for every real $x \neq 0$.

3. (10 points) Suppose

$$F(T) = 1 + a_1T + a_2T^2 + \cdots$$

has positive radius of convergence and every a_k is a positive real number. The formal inverse $G(T) = \sum_{n \geq 0} b_n T^n \in \mathbb{C}[[T]]$ satisfies $F(T)G(T) = 1$. Prove that G also has positive radius of convergence.

Hint. First find a recurrence for b_n , then choose $0 < \delta < 1$ such that $\sum_{k \geq 1} a_k \delta^k < 1$.

Solution. Comparison of coefficients gives $b_0 = 1$ and, for $n \geq 1$,

$$b_n = - \sum_{k=1}^n a_k b_{n-k}.$$

Since $\sum_{k \geq 1} a_k T^k$ converges near 0 and vanishes there, choose $\delta > 0$ within its radius so that $S := \sum_{k \geq 1} a_k \delta^k < 1$. We claim $|b_n| \leq \delta^{-n}$. It holds for $n = 0$. If it holds below n , then

$$|b_n| \leq \sum_{k=1}^n a_k \delta^{-(n-k)} = \delta^{-n} \sum_{k=1}^n a_k \delta^k \leq S \delta^{-n} < \delta^{-n}.$$

Thus $\sum b_n T^n$ converges absolutely for $|T| < \delta$, so its radius of convergence is at least $\delta > 0$.